

Robust Low-Rank Prediction Across Single-Cell Multiomics

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Abstract This report addresses the two prediction tasks in the Open Problems Multimodal Single Cell Integration competition: predicting surface protein abundance from gene expression in CITEseq data and predicting gene expression from chromatin accessibility in Multiome data. I treated the task as cross-modality regression under strong donor and temporal distribution shifts. For CITEseq, I used a 128 dimensional truncated singular value decomposition representation, cell-type indicators, and Ridge regression. For Multiome, I compressed ATAC inputs to 128 dimensions, represented normalized RNA targets with 256 latent components, and trained a small multilayer perceptron with a five-seed ensemble. Model choices were based on donor-held-out, time-held-out, and joint donor and time validation rather than a random split. The final submission achieved public and private scores of 0.803684 and 0.751565. The private score was 0.023104 below the strongest fully open-source competition solution. The experiments indicate that temporal transfer, information discarded by low-rank compression, and the mismatch between latent mean squared error and the row-wise correlation metric account for much of the remaining gap.

Problem Formulation

Multimodal single cell assays measure different molecular layers in the same cell. In the first task, CITEseq provides gene expression measurements and asks for the abundance of 140 surface proteins. In the second, Multiome provides chromatin accessibility measurements and asks for expression values for 23,418 genes. The two mappings reflect the flow from accessible DNA to RNA and from RNA to protein, but they are prediction problems rather than direct causal models. Regulatory delays, feedback, technical noise, and incomplete molecular sampling prevent either output modality from being determined exactly by the input modality.

The computational challenge is also unusual. Each cell contains a sparse and noisy vector, the feature spaces are much larger than those used in ordinary tabular regression, and the test distribution includes donors and differentiation times not represented in labelled training data. A model can therefore score well on randomly held out cells while learning donor or batch patterns that fail on future biological states. I designed the pipeline around this distribution shift. Low rank representations controlled memory and noise, while grouped validation determined whether additional complexity improved transfer rather than merely fitting the training distribution.

Data Evaluation and Validation

Kaggle evaluates each cell by the Pearson correlation between its predicted and observed output vector, then averages the cell level correlations (Kaggle, 2022a). This metric rewards the relative expression pattern within a cell and is invariant to positive affine rescaling of a complete prediction vector. It does not directly reward accurate absolute magnitude. This property motivated row normalization of the Multiome targets and also explains why losses based on correlation or cosine similarity can outperform ordinary squared error even when both produce numerically plausible values.

Dataset dimensions and temporal coverage of the training and testing data are checked and listed (Table 1). I used three validation settings to evaluate the model performance. Donor only validation held out one donor at observed days. Time only validation trained on earlier days and

evaluated the latest labelled day. The strict joint setting held out both a donor and the latest labelled time. This was harder than random cross validation and better represented the private leaderboard, which competition writeups describe as containing both an unseen donor and unseen future days [3, 5]. Hyperparameters were selected for stable performance across these settings rather than for the highest result on one convenient split.

Task	Train cells	Test cells	Input	Target	Latest train and test days
CITEseq	70,988	48,663	22,050 genes	140 proteins	Train day 4 Test day 7
Multiome	105,942	55,935	228,942 ATAC peaks	23,418 genes	Train day 7 Test day 10

Table 1 Dataset dimensions and temporal coverage

Models

CITEseq model

The CITEseq input matrix was reduced from 22,050 genes to 128 components with randomized Truncated SVD. The components were standardized and concatenated with seven one-hot cell-type indicators, producing 135 features per cell. A multi-output Ridge model with alpha equal to 1000 predicted all 140 proteins simultaneously. I excluded donor and day identifiers from the final model. Donor is unavailable for genuinely new individuals, while a categorical day feature cannot represent an unseen future day. Cell type was retained because it is biologically relevant and produced a small but consistent improvement across strict folds.

Ridge regression was a deliberate final model rather than only a baseline. The SVD components are correlated summaries of gene programs, and L2 regularization stabilizes their coefficients under donor shift. The CITE mapping also has a relatively low-dimensional output. In this setting a regularized linear map provided strong validation performance without the additional variance and tuning cost of a neural network.

Multiome model

Multiome required stronger compression because each ATAC vector contains 228,942 features and each output contains 23,418 genes. I fitted a 128-component Truncated SVD representation to the combined labelled and unlabeled ATAC inputs; no test targets were used. This label-free transductive step reduced the sparse input to a tractable dense matrix. Each RNA target vector was L2 normalized, compressed to 256 components, and standardized. Target normalization was compatible with the row-wise correlation metric, which depends on expression pattern more than global scale.

The prediction network contained two hidden layers of 256 units with Swish activation and dropout of 0.10, followed by a linear 256 dimensional latent output. It was trained with Adam at a learning rate of 0.001, batch size 1,024, mean squared error, and 21 epochs. I trained five models with seeds 42, 123, 2026, 7, and 2027. Their latent predictions were averaged before inverse standardization and inverse SVD. Negative reconstructed expression values were clipped to zero. A Ridge blend was tested but omitted because the five-seed neural ensemble scored slightly higher.

Scalable prediction

The final Multiome test output contains about 1.31 billion cell-gene values. Materializing the complete prediction as a labelled table would exceed practical notebook memory. I therefore retained the compact latent predictions, reconstructed gene blocks in chunks, clipped and flattened only the required evaluation entries, and merged them with the CITEseq predictions through the official evaluation identifiers. This procedure produced 1,316,698,650 submission rows while keeping the trained models and intermediate arrays reproducible.

Results and model selection

CITEseq parameters: components = 128 and ridge $\alpha = 1000$

The temporal validation scores for 64, 128, and 256 input components were 0.876876, 0.876998, and 0.876896. The nearly flat curve showed that 128 components captured the useful low-rank signal and that doubling the representation did not improve transfer. For the regularization parameter, alpha equal to 1000 achieved a donor-held-out mean of 0.881598 $\pm$ 0.002706, compared with 0.880340 $\pm$ 0.001712 for alpha equal to 10000. The stronger penalty improved the temporal score by only 0.000319 but reduced the donor mean by 0.001258, so alpha equal to 1000 offered the better overall balance. Adding cell-type increased strict validation by 0.000303 $\pm$ 0.000203. Although small, the gain was consistent and justified retaining the feature.

Multiome parameters: 128 input SVD components and a five-seed MLP ensemble

A gene-mean predictor scored 0.560037, while the strict Ridge model reached about 0.5944. Increasing the ATAC input representation from 128 to 256 components changed the Ridge score from 0.594389 to 0.594472, a gain of only 0.000082 for roughly twice the dense representation. I therefore retained 128 components. Explicit day information was rejected after it reduced time-held-out Ridge performance from 0.601190 to 0.590148. The result suggests that day acted as a batch shortcut rather than a transferable descriptor of differentiation state.

All five neural models outperformed the Ridge reference. Their mean was 0.596788 with a standard deviation of 0.000363, and the latent ensemble reached 0.597468, improving on Ridge by 0.003087. A 0.9 MLP and 0.1 Ridge blend reached 0.597403 and was therefore not used in the final pipeline. The modest neural gain indicates that the compressed ATAC to RNA map is largely smooth and low-dimensional, with some nonlinear structure. The small spread across seeds also shows that further gains require a better representation or objective more than additional repetitions of the same architecture.

Leaderboard performance

Submission	Public score	Private score	Private gap from this work
This work	0.803684	0.751565	0
Best solution	Public rank 1	0.774669	0.023104

Table 2 Performance comparison with the public best score

The public-to-private decrease was 0.052119, or 6.49 percent relative to the public score (Table 2, Kaggle, 2022b). This difference is consistent with the harder private distribution, but it cannot be attributed to temporal shift alone. Repeated decisions based on the public leaderboard may also contribute. Because Kaggle reports one combined score, the leaderboard does not identify how

much of the decrease came from CITEseq versus Multiome. The local validation results nevertheless point to Multiome temporal transfer as a major weakness.

Comparison with the Best Publicly Available Model

The first-place solution is the appropriate primary public comparator (Suzuki, 2022a, 2022b). It achieved a private score of 0.775690, exceeding this work by 0.024125. The winning method used task-specific preprocessing followed by a hierarchical MLP encoder–decoder. For CITE-seq, it combined an SVD representation of normalized and imputed gene expression with genes selected according to stable protein correlation across batches and membership in relevant Reactome pathways. For Multiome, it combined separate SVD representations of normalized ATAC values and binarized accessibility patterns. In both tasks, the targets were normalized and represented in an SVD latent space. Multiple decoder stages generated intermediate predictions, while the Multiome model additionally learned a gene-space residual to correct errors remaining after target reconstruction. Training combined a correlation-based objective with latent-space MAE or MSE terms. The final submission blended models trained with different random seeds and models fine-tuned on selected donor, sex, or day subsets (Suzuki, 2022a, 2022b).

The performance gap is therefore better interpreted as a difference in representation, objective design, and ensemble diversity rather than simply network depth. My pipeline compresses each input modality through a single global SVD and uses one compact predictor family. This design was computationally efficient and could be evaluated systematically under strict donor and temporal shifts, but it may discard low-variance genes or ATAC peaks that remain strongly informative for the target modality. In contrast, the winning solution retained complementary continuous and binary input representations, incorporated pathway-informed CITE-seq features, and corrected low-rank Multiome reconstruction through gene-level residual prediction. Its correlation-based loss also optimized an objective more closely aligned with the Kaggle metric, whereas my latent-space MSE only approximated that objective. Finally, the winning ensemble varied both model initialization and training distribution, while my five-seed ensemble primarily reduced optimization variance. These differences provide plausible mechanisms for its 0.024125 advantage, although the combined leaderboard score does not reveal how much of the gap arose from each task.

Critical analysis

Temporal generalization is the central limitation. In Multiome, the donor only Ridge score was 0.663116, but the time only score fell to 0.601190 and the joint score to about 0.5944. Competition writeups similarly report that public evaluation contained no unseen days, while private evaluation included future days (NocturneBflat et al., 2022). The public-to-private drop is therefore consistent with a model that transfers across donors better than across developmental time. It is not sufficient evidence of ordinary overfitting by itself.

The target SVD oracle provides a second diagnostic. On the strict split, reconstructing the true target through the same 256 dimensional target representation scored 0.629783. The best single MLP scored 0.597377, leaving about 0.0324 between the predictor and this representation-specific ceiling, while the oracle itself remained well below perfect correlation. Error therefore arose from both cross-modality prediction and lossy target compression. A hybrid output that combines latent predictions with selected gene-level residuals could address both sources.

The training objective was also imperfectly aligned with evaluation. Mean squared error penalizes absolute latent deviations, while Kaggle scores relative within cell patterns. Future work should train with a row centered correlation or cosine loss, test feature selection separately within each donor day group, and combine a stable raw feature branch with the SVD branch. Model diversity should come from different representations and losses rather than more random seeds alone. Finally, the input SVD fitted to labelled and unlabelled test inputs was transductive and used no target information, but a deployment study should also report a training-only representation to measure performance on a completely new cohort.

Conclusion

This project produced a complete and computationally feasible pipeline for two large single cell cross-modality tasks. Strict validation supported a regularized linear CITEseq model and a small nonlinear Multiome ensemble, while several plausible additions, including larger input SVDs, day features, and a Ridge blend, were rejected after controlled tests. The final private score of 0.751565 was 0.023104 below the strongest fully open-source solution. The remaining gap is best explained by temporal distribution shift, low-rank information loss, metric mismatch, and limited ensemble diversity. These findings provide concrete directions for improving accuracy without abandoning the pipeline's emphasis on reproducibility and evidence-based model selection.

References

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